

Semantic-Guided Fusion Network for Multi-source Remote Sensing Image Classification

Yuwei Zhao, Chuanzheng Gong, Baogui Huan, Feng Gao, *Member, IEEE*, Junyu Dong, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

Abstract—Multi-source remote sensing image classification has attracted increasing attention due to the complementary spectral, structural, and geometric information. However, existing methods still suffer from two limitations: insufficient semantic contextual modeling and unreliable feature fusion caused by slight spatial misalignment. To address these issues, we propose a Semantic-Guided Fusion Network (SGFNet) for multi-source remote sensing image classification. Specifically, the *Semantic Mixing Convolution Block (SMCB)* is designed to dynamically generate semantic-aware convolution kernels according to contextual relationships among feature representations, thereby enhancing semantic dependency modeling and adaptive feature aggregation. In addition, the *Frequency Modulated Fusion Block (FMFB)* is introduced to perform cross-modal interaction in the frequency domain, which effectively alleviates the influence of slight spatial misalignment and improves complementary information fusion. Extensive experiments conducted on the Augsburg and Houston 2018 datasets demonstrate that the proposed SGFNet consistently outperforms several state-of-the-art methods. The code will be made publicly available at <https://github.com/oucailab/SGFNet>.

Index Terms—Multi-source data classification; Hyperspectral image; Light detection and ranging; Synthetic aperture radar; Multi-source data fusion.

I. INTRODUCTION

WITH the continuous growth of satellite missions and sensor technologies, massive volumes of high-resolution and multi-source remote sensing data are now available, providing unprecedented opportunities to monitor the Earth's surface in a timely and comprehensive manner. Accurately identifying and mapping land-cover types is essential for urban planning, precision agriculture, and disaster management [1].

With the rapid development of Earth observation technologies, a large volume of multi-source remote sensing data has become increasingly accessible. Hyperspectral images (HSIs) capture detailed spectral information across hundreds of contiguous wavelength bands. This rich spectral resolution makes HSI particularly valuable for land cover classification. However, HSI is susceptible to noise and atmospheric interference, which reduces data quality and reliability. To mitigate these challenges, light detection and ranging (LiDAR) and

synthetic aperture radar (SAR) data are frequently integrated as complementary sources [2]. LiDAR provides accurate, high-resolution 3D geometric and elevation information, while SAR offers valuable structural, scattering, and dielectric properties regardless of weather and lighting conditions. Consequently, the joint classification of HSI and LiDAR/SAR data has emerged as an effective paradigm to boost classification accuracy and robustness [3]. Therefore, in this letter, we mainly focus on HSI and SAR/LiDAR joint classification.

Deep learning-based methods have gained considerable research interest for multi-source remote sensing image joint classification, due to their powerful capability in learning hierarchical feature representations and modeling complex nonlinear relationships [4]. By leveraging convolutional neural networks (CNNs) [5] [6], Transformers [7] [8], and more recently Mamba architectures [9], these methods can effectively extract discriminative spatial-spectral features from HSI while simultaneously capturing complementary structural and geometric information from SAR and LiDAR data.

Although existing methods have achieved remarkable classification performance, they still suffer from two limitations: **1) Lacking adaptive semantic and contextual representation capability.** Traditional convolutions use spatially shared and fixed kernels for all spatial locations. Such fixed kernels can hardly adapt to varying semantic structures and content-aware feature distributions. As a result, existing CNN-based methods often fail to effectively capture long-range semantic correlations in complex scenes. **2) Slight cross-modal spatial misalignment weakens reliable feature fusion.** Multi-source remote sensing data acquired by different sensors often suffer from slight spatial misalignment due to inconsistent imaging geometries and sensor characteristics. Although the positional deviations are usually small, local regions and object boundaries may still exhibit structural inconsistencies across modalities. Such misalignment makes direct spatial feature fusion unreliable.

To effectively overcome these limitations, we propose the **Semantic-Guided Fusion Network (SGFNet)** for multi-source data joint classification. Specifically, to enhance the semantic and contextual feature representations for HSI and SAR/LiDAR data, we propose the *Semantic Mixing Convolution Block (SMCB)*. It dynamically generates semantic-aware convolution kernels according to contextual relationships among feature representations. By enabling position-specific contextual modeling and selectively emphasizing semantically relevant information, the proposed module effectively enhances global semantic dependencies. In addition, we introduce the *Frequency Modulated Fusion Block (FMFB)*

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Yuwei Zhao, Chuanzheng Gong, Baogui Huan, Feng Gao, and Junyu Dong are with the State Key Laboratory of Physical Oceanography, Ocean University of China, Qingdao 266100, China. (Email: gaofeng@ouc.edu.cn)

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA.

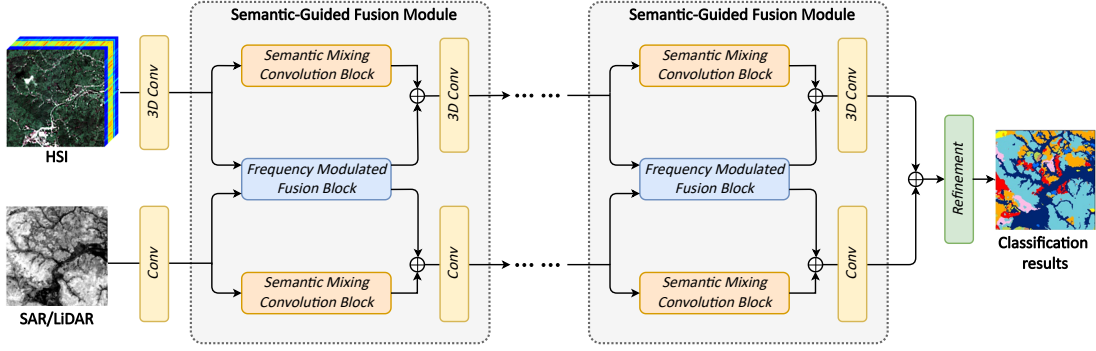


Fig. 1. Overview of the Semantic-Guided Fusion Network (SGFNet). It adopts a dual-branch hierarchical architecture composed of multiple stacked Semantic-Guided Fusion Modules (SGFMs). Within each SGFM, two Semantic Mixing Convolution Blocks (SMCB) are introduced to model semantic-aware contextual representations. These blocks enhance feature interaction by dynamically integrating global semantic context into local convolution. To further strengthen cross-modal interaction, a Frequency Modulated Fusion Block (FMFB) is inserted between the two branches. It performs frequency-aware feature modulation and adaptive information exchange across modalities.

to alleviate the slight spatial misalignment between multi-source data. It performs cross-modal feature interaction in the frequency domain rather than directly conducting spatial-domain fusion. Through adaptive frequency-aware modulation, our FMFB effectively enhances complementary information integration and improves the discriminative capability of multi-source feature representations.

Our main contributions are summarized as follows:

- **Feature Enhancement.** We propose SMCB to enhance semantic and contextual feature representations. Unlike conventional convolutions with spatially shared fixed kernels, our SMCB uses semantic-aware dynamic convolution to improve contextual dependency modeling.
- **Cross-Modal Interaction.** We introduce FMFB to alleviate the slight spatial misalignment. By performing adaptive cross-modal interaction in the frequency domain, our FMFB enhances complementary information integration.
- **Experimental Contribution.** Extensive experiments conducted on two benchmark datasets demonstrate that the proposed SGFNet consistently outperforms several state-of-the-art methods. In addition, the source codes will be publicly released to facilitate future research on multi-source data fusion.

II. METHODOLOGY

A. Overall Framework of the Proposed SGFNet

The overall framework of the proposed Semantic-Guided Fusion Network (SGFNet) is illustrated in Fig. 1. The framework adopts a dual-branch architecture, where the HSI and SAR/LiDAR data are first processed separately through convolutional layers to extract preliminary spectral-spatial and structural representations. To achieve effective cross-modal interaction, the network introduces multiple stacked Semantic-Guided Fusion Modules (SGFM). Within each module, a Semantic Mixing Convolution Block (SMCB) is employed to capture high-level semantic contextual information and enhance discriminative feature learning. Meanwhile, a Frequency Modulated Fusion Block (FMFB) is designed to model complementary frequency-domain characteristics between heterogeneous modalities, enabling the network to preserve fine-

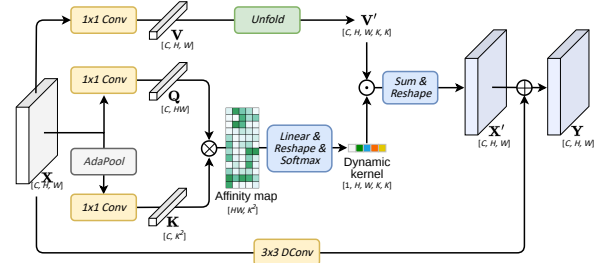


Fig. 2. Illustration of the Semantic Mixing Convolution Block (SMCB).

grained structural details and suppress redundant or noisy information. The semantic and frequency-aware features are then adaptively aggregated through residual fusion operations, promoting more robust cross-modal feature interaction.

By progressively stacking several SGFMs, the framework performs hierarchical semantic refinement and deep multi-modal feature fusion. Finally, the fused representations are integrated through two convolutional layers for refinement to further enhance category consistency and discrimination, producing the final classification results.

B. Semantic Mixing Convolution Block (SMCB)

Standard convolutions use spatially shared fixed kernels, limiting their ability to adapt to varying semantic structures and contextual relationships. To address this issue, we propose the Semantic Mixing Convolution Block (SMCB), which integrates convolution with semantic-aware dynamic filtering. Instead of fixed kernels, SMCB dynamically generates spatially adaptive kernels based on semantic affinities between query and key features. The resulting dynamic kernels guide local feature aggregation, enabling the network to emphasize semantically relevant contextual information adaptively. Details of the SMCB are shown in Fig. 2. Given an input feature $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, it is projected by a 1×1 convolution to obtain the value and query features as:

$$\mathbf{V} = f_v(\mathbf{X}), \quad \mathbf{Q} = f_q(\mathbf{X}), \quad (1)$$

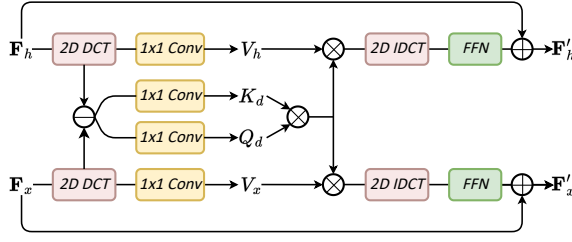


Fig. 3. Illustration of the Frequency Modulated Fusion Block (FMFB).

where f_v and f_q denote the convolution layer. Meanwhile, $\mathbf{X}$ is passed through an adaptive pooling layer to extract compact contextual information as:

$$\mathbf{K} = f_k(\text{AdaPool}(\mathbf{X})), \quad (2)$$

where f_k denotes the convolution layer. We obtain $\mathbf{Q} \in \mathbb{R}^{C \times HW}$ and $\mathbf{K} \in \mathbb{R}^{C \times K^2}$ via reshaping. Here K denotes the dynamic kernel size. The query and key features are multiplied to obtain an affinity map:

$$\mathbf{A} = \mathbf{Q}^T \mathbf{K}. \quad (3)$$

This affinity map measures the relationship between each spatial position and the K^2 sampling locations in a local window. It is then processed by a linear layer, reshaped, and normalized by Softmax to generate the dynamic convolution kernel:

$$\mathbf{D} = \text{Softmax}(\text{Reshape}(\text{Linear}(\mathbf{A}))). \quad (4)$$

In parallel, the value feature $\mathbf{V}$ is unfolded into local patches:

$$\mathbf{V}' = \text{Unfold}(\mathbf{V}). \quad (5)$$

The dynamic kernel $\mathbf{D}$ is then applied to the unfolded value feature. In each channel of $\mathbf{V}'$, for every spatial position (h, w) , the output feature is computed by weighted summation over its local $K \times K$ neighborhood as:

$$\mathbf{X}' = \sum_{i=1}^K \sum_{j=1}^K \mathbf{D}_{h,w,i,j} \cdot \mathbf{V}'_{h,w,i,j}. \quad (6)$$

Besides the dynamic context branch, the framework also includes a local feature enhancement branch. The original input $\mathbf{X}$ is processed by a 3×3 depth-wise convolution to compute $\mathbf{X}_d$. Finally, the dynamically aggregated feature $\mathbf{X}'$ is combined with the local depth-wise convolution feature through residual addition to produce the final output:

$$\mathbf{Y} = \mathbf{X}' + \mathbf{X}_d. \quad (7)$$

The SMCB achieves a more flexible and semantically aware feature extraction mechanism. It is particularly suitable for understanding the complex structures of remote sensing data, where both local structural details and semantic contextual dependencies are critical.

C. Frequency Modulated Fusion Block

Multi-source data are usually collected by different sensors with varying imaging geometries and acquisition times, which often results in slight spatial misalignment between modalities. Local object regions may exhibit small positional deviations and structural inconsistencies. Such slight misalignment may weaken the effectiveness of cross-modal feature fusion. To this end, we propose the Frequency Modulated Fusion Block (FMFB), which performs adaptive cross-modal fusion in the frequency domain to enhance feature consistency.

Details of FMFB are illustrated in Fig. 3. It takes the feature representations $\mathbf{F}_h$ from the HSI branch and $\mathbf{F}_x$ from the SAR/LiDAR branch as inputs. It first applies 2D Discrete Cosine Transform (DCT) to project both features into the frequency domain:

$$\hat{\mathbf{F}}_h = \text{DCT}(\mathbf{F}_h), \quad \hat{\mathbf{F}}_x = \text{DCT}(\mathbf{F}_x). \quad (8)$$

We then compute the element-wise difference in the frequency domain:

$$\hat{\mathbf{F}}_d = \hat{\mathbf{F}}_h - \hat{\mathbf{F}}_x. \quad (9)$$

Based on this joint representation, two 1×1 convolution layers generate the query and key features as $\mathbf{Q}_d = f_q(\hat{\mathbf{F}}_d)$, $\mathbf{K}_d = f_k(\hat{\mathbf{F}}_d)$.

Meanwhile, the two DCT-transformed inputs are separately projected into value features:

$$\mathbf{V}_h = f_v^h(\hat{\mathbf{F}}_h), \quad \mathbf{V}_x = f_v^x(\hat{\mathbf{F}}_x). \quad (10)$$

Next, attention is computed as follows:

$$\mathbf{F}_h^{\text{attn}} = \text{Softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_d}{\sqrt{d_k}} \mathbf{V}_h\right), \quad (11)$$

$$\mathbf{F}_x^{\text{attn}} = \text{Softmax}\left(\frac{\mathbf{Q}_d \mathbf{K}_d}{\sqrt{d_k}} \mathbf{V}_x\right), \quad (12)$$

After that, the modulated frequency-domain features are transformed back into the spatial domain through 2D inverse DCT:

$$\hat{\mathbf{F}}'_h = \text{IDCT}(\mathbf{F}_h^{\text{attn}}), \quad \hat{\mathbf{F}}'_x = \text{IDCT}(\mathbf{F}_x^{\text{attn}}). \quad (13)$$

The reconstructed feature is further refined by the Feed-Forward Network (FFN) and residual connections:

$$\mathbf{F}'_h = \text{FFN}(\hat{\mathbf{F}}'_h) + \mathbf{F}_h, \quad \mathbf{F}'_x = \text{FFN}(\hat{\mathbf{F}}'_x) + \mathbf{F}_x. \quad (14)$$

III. EXPERIMENTAL RESULTS AND ANALYSIS

A. Datasets and Experimental Settings

The performance of the proposed SGFNet is evaluated on two benchmark multi-source remote sensing datasets for land-cover classification. The first dataset is the Augsburg Dataset [10], which consists of HSI and SAR data. It contains ground-truth annotations for seven land-cover classes and has a spatial size of 332×485 pixels. The second dataset is the Houston 2018 Dataset, which comprises HSI and LiDAR data. As it is obtained from the IEEE GRSS Data Fusion Contest 2018 [11], this dataset provides a representative multi-sensor optical geospatial benchmark for challenging urban land-cover and land-use classification tasks over the University of Houston

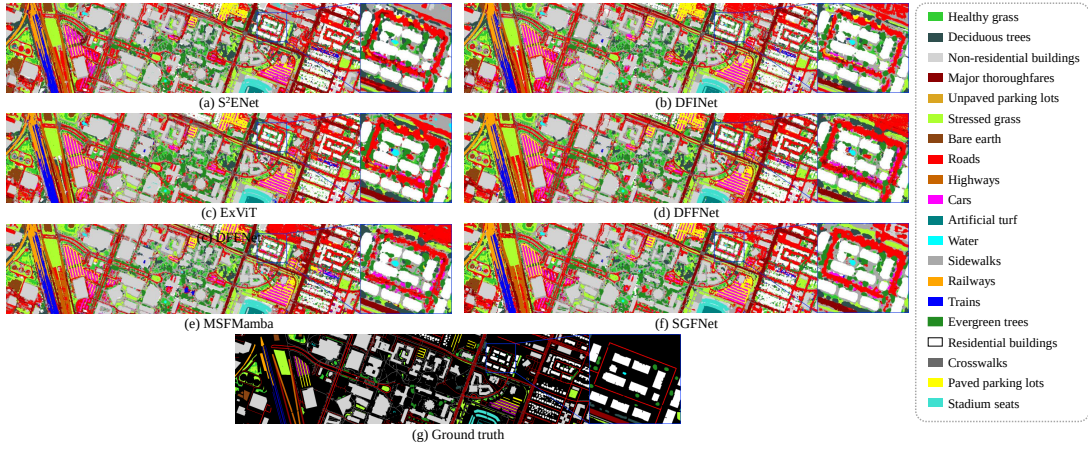


Fig. 4. Classification results of different methods on the Houston2018 dataset.

TABLE I
EXPERIMENTAL RESULTS ON THE AUGSBURG DATASET. THE **BOLD** AND UNDERLINE DENOTE THE BEST AND SECOND RESULTS.

Class	S ² ENet	DFINet	ExViT	DFFNet	MSFM	SGFNet
Forest	98.10	97.38	90.04	96.17	97.17	97.18
Residential area	99.08	98.37	95.44	96.64	98.15	98.28
Industrial area	12.19	<u>61.31</u>	34.58	37.70	50.26	61.41
Low plants	91.78	92.63	90.68	94.21	95.52	<u>95.48</u>
Allotment	45.12	49.33	51.82	<u>53.54</u>	53.35	65.01
Commercial area	1.22	3.54	28.63	<u>11.36</u>	2.63	8.55
Water	24.09	26.61	17.65	27.15	<u>49.97</u>	56.62
OA	88.22	90.66	86.65	89.37	<u>91.38</u>	92.38
AA	53.08	61.31	58.41	59.54	<u>63.31</u>	68.93
Kappa	86.47	86.47	80.79	84.51	<u>87.45</u>	89.05

campus and its surrounding areas. The dataset includes hyper-spectral imagery with 48 spectral bands spanning wavelengths from 380-1050 nm, along with LiDAR measurements that provide accurate elevation information.

Our SGFNet was conducted on NVIDIA GeForce RTX 5080 GPU. The training phase spanned over 100 epochs. The AdamW optimizer was used with a learning rate of 0.001. The batch size was set to 32. To evaluate the performance of the proposed SGFNet, we compare it against other state-of-the-art methods, including S²ENet [12], DFINet [13], ExViT [14], DFFNet [15], and MSFM [16]. To quantitatively evaluate the experimental results, we employ three standard metrics: Overall Accuracy (OA), Average Accuracy (AA), and Kappa coefficient.

B. Parameter Analysis

Number of SGFM. The number of SGFM is an important parameter that may affect the classification performance. We test different numbers of SGFM from 1 to 4. The experimental results are shown in Fig. 5(a). When the number of SGFM is set to 2, our SGFNet achieves the best performance on both datasets. Therefore, in the following experiments, we use two SGFMs in the proposed SGFNet.

Number of Principal Components for HSI. HSI data contains rich spectral information but also introduces redundancy and spectral noise. Therefore, we use Principal Component

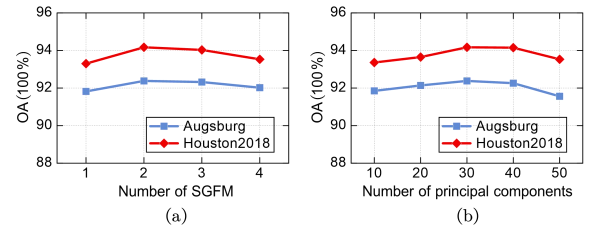


Fig. 5. Parameter analysis. (a) The relationship between OA and the number of SGFM. (b) The relationship between OA and the number of principal components after PCA for HSI.

Analysis (PCA) for HSI preprocessing. We tested the number of principal components for HSI, and the experimental results are shown in Fig. 5(b). It can be observed that when the number of principal components is set to 30, our SGFNet achieves the best classification performance. Retaining too few components discards crucial discriminative spectral features, leading to suboptimal accuracy. Thus, the number of principal components is empirically set to 30.

C. Experimental Results and Analysis

Results on the Augsburg Dataset. Table I shows the quantitative evaluations on the Augsburg dataset. The OA value of the proposed SGFNet achieves 92.38%, outperforming all the other methods. From the class-wise results, SGFNet achieves the best performance in the Industrial area and Allotment. Both categories usually exhibit complex spatial structures and significant spectral ambiguity, indicating that SGFNet can effectively capture discriminative semantic and contextual information.

Results on the Houston2018 Dataset. The classification maps on the Houston2018 dataset are shown in Fig. 4, and the corresponding quantitative evaluations are illustrated in Table II. It can be observed that SGFNet produces classification maps that are more consistent with the ground truth, demonstrating superior land-cover discrimination capability. Table II shows that our SGFNet obtains the highest OA and Kappa, significantly outperforming other state-of-the-art methods. SGFNet achieves the best performance in several

TABLE II
EXPERIMENTAL RESULTS ON THE HOUSTON2018 DATASET. THE **BOLD**
AND UNDERLINE DENOTE THE BEST AND SECOND BEST RESULTS.

Class	S ² ENet	DFINet	ExViT	DFFNet	MSFM	SGFNet
Healthy grass	99.15	<u>94.29</u>	93.34	92.63	90.38	92.07
Stressed grass	84.35	92.71	<u>93.44</u>	92.44	94.04	93.18
Artificial turf	100.0	100.0	100.0	100.0	100.0	100.0
Evergreen trees	98.13	98.92	95.87	<u>99.38</u>	99.34	99.76
Deciduous trees	86.39	97.77	96.85	<u>99.16</u>	99.11	99.94
Bare earth	99.33	100.0	100.0	<u>99.96</u>	100.0	100.0
Water	100.0	100.0	100.0	100.0	100.0	100.0
Residential buildings	97.17	97.36	97.39	<u>98.00</u>	96.27	98.78
Non-residential buildings	93.93	93.29	93.00	93.73	<u>95.58</u>	96.24
Roads	70.17	75.73	72.18	<u>80.31</u>	75.90	85.33
Sidewalks	72.11	<u>83.67</u>	69.69	72.59	81.43	84.64
Crosswalks	82.47	94.23	88.12	91.84	<u>96.22</u>	98.14
Major thoroughfares	89.44	81.20	86.12	84.54	86.89	<u>88.85</u>
Highways	98.53	98.91	99.50	98.82	98.63	<u>99.35</u>
Railways	99.21	99.94	99.87	<u>99.89</u>	99.69	99.82
Paved parking lots	95.62	98.37	97.02	97.78	<u>98.94</u>	99.34
Unpaved parking lots	100.0	100.0	100.0	100.0	100.0	100.0
Cars	96.77	99.09	98.13	<u>98.89</u>	97.81	98.67
Trains	100.0	99.46	<u>99.98</u>	99.91	99.97	100.0
Stadium seats	100.0	99.98	<u>99.99</u>	100.0	100.0	100.0
OA	90.05	91.02	89.98	91.21	<u>92.38</u>	94.17
AA	93.14	95.24	94.02	94.99	<u>95.51</u>	96.71
Kappa	87.20	88.46	87.14	88.68	<u>90.16</u>	92.46

TABLE III
ABLATION STUDY OF THE PROPOSED SGFNET.

Model Variant	Augsburg	Houston2018
Baseline	88.54	89.65
w/o SMCB	89.88	91.01
w/o FMFB	91.29	92.37
SGFNet (Full model)	92.38	94.17

challenging categories, including Evergreen trees, Deciduous trees, Buildings, Roads, Sidewalks, Crosswalks, and Paved parking lots. These categories usually contain complex spatial distributions and strong spectral similarity with surrounding objects. The superior performance of SGFNet indicates that the proposed framework can effectively capture discriminative semantic information and exploit complementary features from multi-source data.

D. Ablation Study

Table III presents the ablation study results of the proposed SGFNet. The experiments are designed to evaluate the effectiveness of SMCB and FMFB. Starting from the baseline model, both modules consistently improve the classification performance. This demonstrates that SMCB can effectively enhance contextual representation capability, and FMFB effectively alleviates the slight spatial misalignment between multi-source data. The complete SGFNet achieves significant improvements of 3.84% and 4.52%, respectively. These results demonstrate that both proposed modules are complementary and jointly contribute to more robust multi-source feature fusion and classification.

IV. CONCLUSIONS

In this letter, we propose SGFNet for multi-source remote sensing image classification. To address the limitation

of traditional convolutions in modeling semantic contextual relationships, we propose Semantic Mixing Convolution Block (SMCB) to dynamically generate semantic-aware convolution kernels according to contextual affinities among feature representations. In addition, we introduce the Frequency Modulated Fusion Block (FMFB) to perform adaptive cross-modal interaction in the frequency domain. Extensive experiments on two benchmark datasets demonstrate that SGFNet consistently outperforms state-of-the-art methods.

REFERENCES

- [1] P. Duan, S. Jin, X. Lu, L. Liang, X. Kang, and A. Plaza, "Domain-adaptive mamba for cross-scene hyperspectral image classification," *IEEE Transactions on Image Processing*, vol. 35, pp. 1206–1217, 2026.
- [2] X. He, K. Xu, S. Wang, P. Zhao, Y. Zhang, and J. Bi, "Mufnet: A mamba-based uncertainty-aware fusion framework for fine-grained satellite hyperspectral-lidar tree species mapping," *IEEE Geoscience and Remote Sensing Letters*, vol. 23, pp. 1–5, 2026.
- [3] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "MSFMamba: Multiscale feature fusion state space model for multi-source remote sensing image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–16, 2025.
- [4] X. Liu, X. Tang, T. Lu, and K. Ding, "HSI-LiDAR joint classification via progressive spatial-spectral-frequency fusion learning," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, pp. 1–12, 2026.
- [5] K. Li, D. Wang, X. Wang, G. Liu, Z. Wu, and Q. Wang, "Mixing self-attention and convolution: A unified framework for multisource remote sensing data classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–16, 2023.
- [6] C. Gong, F. Gao, J. Lin, J. Dong, and Q. Du, "Representative spectral correlation network for multisource remote sensing image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 64, pp. 1–14, 2026.
- [7] Y. Zhao, F. Gao, X. Jin, J. Dong, and Q. Du, "Dynamic frequency feature fusion network for multisource remote sensing data classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.
- [8] X. Cui and L. Zhang, "MTMixer: a hybrid Mamba-Transformer architecture for multimodal remote sensing image classification," *International Journal of Remote Sensing*, vol. 47, no. 2, pp. 770–799, 2026.
- [9] Z. Meng, Z. Yang, and F. Zhao, "Joint classification of hyperspectral and lidar data based on dynamic sparse-dense transformer and mamba," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 19, pp. 11445–11461, 2026.
- [10] J. Hu, R. Liu, D. Hong, A. Camero, J. Yao, M. Schneider, F. Kurz, K. Segl, and X. X. Zhu, "Mdas: a new multimodal benchmark dataset for remote sensing," *Earth System Science Data*, vol. 15, no. 1, pp. 113–131, 2023.
- [11] Y. Xu, B. Du, L. Zhang, D. Cerra, M. Pato, E. Carmona, S. Prasad, N. Yokoya, R. Hänsch, and B. Le Saux, "Advanced multi-sensor optical remote sensing for urban land use and land cover classification: Outcome of the 2018 ieec grss data fusion contest," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 12, no. 6, pp. 1709–1724, 2019.
- [12] S. Fang, K. Li, and Z. Li, "S²ENet: Spatial-spectral cross-modal enhancement network for classification of hyperspectral and LiDAR data," *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [13] Y. Gao, W. Li, M. Zhang, J. Wang, W. Sun, R. Tao, and Q. Du, "Hyperspectral and multispectral classification for coastal wetland using depthwise feature interaction network," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–15, 2022.
- [14] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (exvit) for land use and land cover classification: A multimodal deep learning framework," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–15, 2023.
- [15] Y. Zhao, F. Gao, X. Jin, J. Dong, and Q. Du, "Dynamic frequency feature fusion network for multisource remote sensing data classification," *IEEE Geoscience and Remote Sensing Letters*, vol. 22, pp. 1–5, 2025.
- [16] F. Gao, X. Jin, X. Zhou, J. Dong, and Q. Du, "Msfmamba: Multi-scale feature fusion state space model for multi-source remote sensing image classification," *IEEE Transactions on Geoscience and Remote Sensing*, 2025.